S.6 GEOGRAPHY SEMINAR ITEMS

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09:00 AM

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ITEM 1

Scientists have observed that the continents are distributed across the earth in different positions, with large oceans separating them. However, several pieces of evidence suggest that some continents were once joined together. For example, the coastlines of Africa and South America appear to fit together, while similar fossils, rock formations and mountain systems have been discovered on continents that are now far apart. Coal deposits have also been found in places where present-day climatic conditions do not favour their formation, suggesting that those areas may have experienced different environmental conditions in the past.

These observations have attracted the attention of geographers because the movement and changing positions of the continents have influenced the physical characteristics of different parts of the world. The distribution of rocks, soils, minerals, fossil fuels and other natural resources is not uniform. Some areas have valuable deposits of petroleum, coal, iron ore, gold, copper and other resources that support industries, employment, trade and improved standards of living.

During a geographical inquiry, your class has been asked to investigate how the present arrangement of the continents can be explained using scientific evidence and how this arrangement is related to the availability of natural resources that people depend on for their livelihoods and economic development.

You are a **geographical researcher** whose role is to examine the available evidence and use a relevant geographical theory to explain the distribution of the continents and its relationship with natural resources.

Task

Write an essay using the Continental Drift Theory to explain the distribution of the continents and show how the changes in their positions contributed to the formation and distribution of natural resources that people can exploit to improve their lives. Support your essay with relevant scientific evidence and examples from Africa and other parts of the world.

ITEM 2

Scenario

A farming community in western Uganda has experienced declining agricultural productivity over several years. Farmers report that some fields no longer produce the same yields as they did in the past, while others experience soil erosion, loss of fertility and poor water retention. In some areas, continuous cultivation, burning of vegetation, overgrazing and cultivation on steep slopes have exposed the soil to degradation. However, farmers have noticed that soils differ from one place to another in terms of colour, texture, depth, drainage and fertility, and these differences influence the types of crops that can grow successfully.

The local community development office has therefore organised a geographical investigation to help farmers understand the characteristics and quality of their soils. Learners have been invited to gather information through observation, simple soil investigations and discussions with farmers. They are expected to analyse the information collected and use it to suggest practical ways of managing soil resources while maintaining agricultural production for future generations.

You are a **soil management adviser** working with the community. Your role is to examine the available information about the soils, identify their important properties and quality, and recommend practical measures that can promote sustainable agricultural production.

Task

Write a geographical report proposing feasible ways of using and managing the soils in the community. In your response, use information about soil properties, formation and quality to explain the recommended practices and show how they can promote sustainable development.

ITEM 3

A district in Uganda has a river system that begins in a highland area and flows through several landscapes before eventually entering a large lake. Along its course, the river changes in character. In the upper section, the river flows rapidly through steep valleys and rocky channels, while in the middle section, the valley becomes wider and the river develops meanders. Further downstream, the river flows more slowly across a broad, relatively flat area where deposition has created fertile floodplains. During periods of heavy rainfall, the river sometimes overflows its banks and floods neighbouring areas.

The river and its associated features have created opportunities for human development. Communities use the water for domestic purposes and irrigation, practise fishing, cultivate fertile floodplain soils, keep livestock and operate small businesses. The waterfalls and attractive river landscapes also have potential for tourism and hydroelectric power generation. However, cultivation close to riverbanks, removal of vegetation, settlement on floodplains, sand extraction and disposal of waste into the river are affecting the river system and surrounding ecosystems.

The district authorities are concerned that increasing exploitation of the river and its resources may cause environmental degradation and increase the risks of flooding, soil erosion and water pollution. They have therefore asked learners to investigate the river system and identify ways in which its features and related resources can be used sustainably for the benefit of the community.

You are a **geographical resource-planning team** working with the district authorities. Your role is to analyze the river system, its features and associated resources and propose sustainable ways of using them while protecting the landscape and ecosystems.

Task

Task

Write a geographical essay proposing sustainable ways of harnessing river features and their related resources for development. In your essay, use knowledge of the river cycle to explain how river features develop from the upper course to the lower course, identify the resources associated with these features, and propose practical and sustainable ways in which communities can utilize them to improve their livelihoods and promote development while conserving the river environment.

ITEM 4

A group of Senior Five learners in Uganda have been observing that weather conditions change from one day to another. On some days, the sky is clear and temperatures are high, while on other days there are clouds, strong winds and heavy rainfall. They have also noticed that aircraft travel at high altitudes, communication signals can travel through the upper atmosphere, and temperatures vary considerably at different heights above the Earth's surface.

During a Geography lesson, the learners learn that the Earth is surrounded by a gaseous envelope that extends several kilometres above its surface. This envelope contains gases in different proportions, together with water vapour and tiny particles. The gases are not distributed equally in all parts of the atmosphere, and atmospheric conditions also change with increasing altitude. Some parts of the atmosphere are associated with clouds and rainfall, while another region contains a gas that absorbs much of the harmful ultraviolet radiation from the Sun.

The learners are interested in understanding why the atmosphere is essential for weather and climate and how conditions in its different layers influence human beings, plants, animals and other forms of life. They therefore decide to investigate the composition

and structure of the atmosphere and examine how its different layers contribute to conditions experienced on Earth.

You are a **Senior Five Geography learner conducting a geographical inquiry**. Your role is to analyze the atmosphere by examining its composition and layers and establish how these characteristics influence weather, climate and life on Earth.

Task

Prepare a geographical report analyzing the atmosphere and its influence on weather, climate and life on Earth. In your report, examine the composition and different layers of the atmosphere and explain how the characteristics of each layer influence weather and climate and support, protect or sustain life on Earth.

ITEM 5

Uganda experiences sunshine during much of the year, although the amount of energy received at the Earth's surface varies from one place to another and from day to day. On sunny days, the ground and buildings become warmer, water evaporates from lakes and rivers, and farmers spread crops such as maize, beans and coffee in open spaces to dry. On cloudy days, temperatures may be lower and agricultural produce takes longer to dry. Places near the Equator generally experience considerable sunshine throughout the year, while differences in altitude, season and length of daylight influence local temperature conditions. The availability of sunshine has important implications for human activities. Farmers require energy to dry and preserve their produce, households need electricity for lighting and cooking, and businesses depend on reliable energy to operate machinery and equipment. Some communities in Uganda experience high electricity costs and unreliable power supplies, while farmers lose part of their harvest because of inadequate storage facilities. However, a number of entrepreneurs have started using solar panels to generate electricity, solar dryers to preserve agricultural products and solar-powered pumps to supply water for irrigation.

The district authorities believe that the sunshine available in the area could provide more opportunities for economic development. They want to understand the conditions that influence its availability and identify ways of turning this resource into commercially valuable products and services. They have therefore invited Geography learners to investigate the energy received from the Sun, examine the factors that influence its distribution and recommend practical ways of using it to improve agriculture, industry, employment and people's standards of living.

You have been selected as a Geography and development adviser to assist the district authorities in identifying opportunities for the productive and sustainable use of this resource.

Task

Prepare a geographical report in which you describe the nature of solar radiation, explain the atmospheric processes associated with it, analyze the factors influencing the amount of solar radiation received at the Earth's surface, and propose practical strategies for commercializing solar radiation to boost agricultural production, industrial development, employment, household incomes and sustainable development in Uganda.

ITEM 6

A rapidly growing town in Uganda is located in a valley surrounded by hills. During some early mornings, residents notice that cold air remains close to the ground while the air above the valley feels warmer. A layer of mist sometimes covers the valley, visibility becomes poor, and smoke from vehicles, factories and household cooking remains concentrated near the ground. People travelling to work experience difficulties seeing clearly, while some residents complain about unpleasant air and breathing discomfort.

The local weather station has also recorded calm winds and clear skies during some of these mornings. In contrast, nearby hilltops sometimes experience warmer conditions

than the valley floor. As the Sun rises higher in the sky, the mist gradually disappears, and the air near the ground becomes warmer. These observations have raised concerns among local authorities about the relationship between temperature conditions at different heights, local weather patterns and air quality.

The district environmental office has invited Geography learners to investigate the unusual temperature conditions and explain why they occur, how they influence the local environment and what measures can reduce their negative effects. The findings will help the authorities improve environmental management and protect the health of residents.

You have been selected as a Geography researcher to investigate the situation and advise the district authorities.

Task

Prepare a geographical report analyzing the causes of the above atmospheric process, its effects on local weather patterns, climate systems and human activities. In your report, also suggest practical measures for reducing its harmful consequences.

ITEM 7

During a Geography fieldwork activity around Lake Victoria and the surrounding areas of Kampala, a group of A-Level learners observed that weather conditions changed from one place to another and at different times of the day. In the early morning, the land was relatively cool, while the lake remained warmer. Later in the day, the land became hotter, and winds began blowing from the lake towards the land. On some afternoons, strong winds developed, followed by cloud formation and rainfall.

The learners also observed that on certain days, the air seemed to rise over the hotter land, while cooler air moved in to replace it. Their teacher explained that differences in the heating of the Earth's surface influence the weight, density and movement of air, resulting in changes in atmospheric pressure and wind direction. The learners were

therefore encouraged to investigate how temperature differences influence atmospheric pressure and how variations in pressure determine wind movement.

As part of their investigation, the learners collected weather information, drew diagrams showing air movement, and compared the weather conditions over the lake and the surrounding land.

Task

Prepare an illustrated fieldwork report describing the atmospheric process responsible for these observations, explaining its causes and analyzing its effects on weather conditions and people's daily activities in Uganda.

ITEM 8

In a mountainous district in eastern Uganda, communities living on hillsides have observed that the land changes differently from one place to another and from one season to another. In some areas, small quantities of soil move slowly downhill, while in other places, rocks fall from steep slopes and large masses of soil suddenly slide into valleys. During periods of prolonged heavy rainfall, some hillsides develop deep gullies, roads become blocked by soil and rocks, and houses located near unstable slopes are threatened.

The situation is different in areas covered by forests, where vegetation protects the soil and reduces the speed of surface runoff. On cultivated hillsides, however, clearing vegetation, digging drainage channels carelessly and growing crops on steep slopes have increased soil erosion and the movement of materials downhill. Some communities experience these problems mainly during rainy seasons, while others face rock falls and soil movement even during relatively dry periods because of weathering, unstable rocks and human activities.

These changes have affected farming, transport, settlement, water quality and people's safety. Farmers have lost fertile soil and crops, roads have been damaged, and some families have been forced to leave dangerous areas. District authorities have therefore

asked Geography learners to investigate why slope processes differ across locations and seasons and recommend ways of reducing the hazards they cause.

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Task

As a Geography researcher, prepare a geographical hazard investigation report analyzing the of slope processes. Examine the factors influencing where and when they occur, assess their impacts on the physical environment and human activities, and propose practical strategies for managing slope-related hazards.